



NURSERY MANAGEMENT AND SEEDLING SELECTION

Planting coconut seed nuts directly in planting holes is not recommended. Properly selected, well grown, 7-10 months old seedlings should be transplanted in the field. This ensures fast growth and a uniform plantation. Seed nuts may be planted either in nursery beds or in poly bags to obtain vigorous seedlings.

Nursery practices

❖ Selection of nursery site

The proposed nursery site should be flat or with a low gradient. The soil should be sandy or sandy loam and well drained. The site should have adequate sunlight, scattered shade and an assured source of water in close proximity (live shade like gliricidia is preferred for nurseries).

❖ Nursery beds (for the production of bear-rooted seedlings)

The length of the bed can be arranged as desired in order to facilitate routine activities. Each bed should accommodate five rows of seed nuts. Seed beds should be about 15-25 cm above ground level and the distance between two rows should be 45 cm. The seed nuts are placed horizontally, with a spacing of 15 cm between two adjacent nuts in a trench, 10-15 cm deep, and covered with a thin layer of soil so that the surface of the nut is just visible. The seed nuts in any single trench should be placed in one direction and in the adjoining trench seed nuts should be placed in the opposite direction. The seed nuts along a trench should also be placed in between two nuts of the adjacent row (figure 1). Seed beds should be separated by shallow drains which join up to form a leader drain to remove the excess water during heavy rains.

❖ Conservation of soil moisture in nursery beds

Normally coconut fronds or coir dust layer with 3-6 cm thickness are used as mulch. If coir dust is used, rotation of mulch materials should be carried out. In areas where fronds or coir dust are not available, weed thrash, straw or saw dust could be used.

❖ Water supply

Regular watering is essential during dry weather. Watering should commence if there is no rain continuously for six days and continue at three day intervals (rainy day = rainfall of 0.25mm).



Figure 1: A demonstrated nursery



Figure 2: mulching nurseries with coconut fronds

❖ Weed control

Weed control is an important practice because weeds compete with coconut seedlings for soil moisture. Weeding should be carried out depending on the weather conditions, once or twice a month.

❖ Pests and diseases

1. Termite control

Termite or white ants cause considerable damage to coconut seed nuts and seedlings (figure 3). Termites are usually attracted to coconut husks and eventually tunnel through the nut damaging the growing point. To prevent this damage, seed nuts should be dipped in a recommended insecticide solution prior to planting. If termites are observed in the soil, and especially in nurseries with clayey soils, drenching of the nursery bed with an insecticide is recommended.



Figure 3: Termite infested coconut seedling

Recommend insecticide/Dosage

Dipping - Marshal 20, 4 ml in one liter of water, dip a nut for 3 minutes prior to planting.

Drenching - Five liters of the above mixture of Marshal 20 is sufficient to drench one square meter. Use watering-can for drenching.

2. Collar rot

Collar rot is common in nursery seedlings with high moisture availability. The basal portion of the seedling starts decaying leading to bacterial infections. Seedling affected with collar rot sometimes show symptoms similar to termite attack but the former could be differentiated due to the characteristic odor emitted from the rotten seed nut. To prevent this situation, deep planting and thick mulching should be avoided. Adequate drainage should be provided in nurseries where the water table is close to the surface.

3. Leaf die-back

In certain instances nursery seedlings show extensive die-back of leaves. The die-back starts from the tip of the leaf and circular brown patches could also be seen in the middle portion of the leaf.



Figure 4: Dieback disease

Curvularia, a mould, causes this condition and is common during periods of dry weather. In nurseries the disease spreads rapidly when plants are regularly watered as it increases the humidity of the environment. This could be easily controlled by spraying 1% copper fungicide available in the market.

❖ Crop rotation

Continuous use of nursery beds reduces the fertility levels of soil. This could be avoided by cultivating leguminous crops, in a short period, on these beds.

❖ Non germinating seed nuts

Non germinated seed nuts should be removed at the end of five months from beds. In a well managed nursery the germination should be at least 80% at the end of five months (figure 5).

❖ Fertilizer application

Fertilizer application is generally not recommended for nurseries. However, application of the following fertilizer mixture is recommended if poly bagged seedlings are kept for long time without planting in the field and they are showing deficiency symptoms. 70 g of the below mixture is applied to each seedling and mix it with the soil. Plants should be irrigated after fertilizer application if there is no rain.

Urea	1.5 parts
Muriate of Potash	2 parts
Triple Super Phosphate	1 parts
Kieserite	1 parts

❖ Seedling selection

Selection of seedlings in the nursery is an important event as planting of a quality seedling is essential to have a vigorous and early bearing palm (figure 6). All non germinated nuts should be removed at the end of five months. The second selection is done after seven months from laying. In a well managed nursery the rejection of both nuts and seedlings should be kept at following percentages.



Figure 5: Well-maintained bareroot seedling nursery

Non germination	8%
Late germination	10%
Weak seedlings	12%
Total	30%

❖ Seedling selection criteria

A vigorous seedling should have the following characters.

- Stout stem – collar region of seedlings
- Dark green leaves
- Broad leaves
- Short petioles
- Sharp ridges of leaves
- Convex leaf surface
- Free from pests and diseases

Seedlings are ready for issue for transplanting from the seventh months from seed nut planting.



Figure 6: Selected bareroot seedling

❖ Raising seedlings in poly bags

Seed nuts are laid in a pre-nursery at vertical position (perianth up ward direction). In a pre-nursery, the distance between adjacent nut rows can be

15cm (6”) and the nuts can be laid with just 5cm (2”) distance between two nuts in a row. When the sprout of seed nuts in the pre-nursery grow up to 5-10 cm, nuts should be transferred to poly bags. The poly bags are made of UV radiation resistant black polythene sleeves of 200-500 gauge, 20/100 mm thick, and 40 x 25 cm in size with gussets measuring 15 cm at two bottom edges. Several small punches should be made on the bag 1” from the bottom to facilitate draining of excess water. With the spacing recommended above, a nursery of one hectare in extent can raise about 50,000 seedlings (20,000 seedlings/ac). Seedlings are suitable for field planting after 8 months of poly bagging, but if necessary, these seedlings can be kept further 4 months in poly bags.



Figure 7: Pre-nursery seedling

❖ **Filling the bags**

The mixture should have,

- Top soil 1 parts
- Cow dung 2 parts
- Coir dust 3 parts

If top soil is available adequately, the above filling mixture could be changed as 3 : 2 : 1.

❖ **Laying in poly bags**

A germinating nut is ready for poly bagging once the sprout is visible well (height of sprout 5 – 10 cm.) (figure 7). Place the sprouted nut upright in the poly bag and fill it with the above soil mixture. The soil mixture is further added so that the upper surface of the nut is barely visible. It is advisable to leave about 3cm from the top edge of the bag to facilitate watering (figure 8).



Figure 8: Correctly polybagged seedling

❖ **Watering**

Application of water should be neither insufficient not excessive. About 2.5 liters/poly bag a day is recommended, preferably in the morning or late afternoon.



Figure 9: Poly bagged seedling nursery

❖ Spacing of bags

Spacing of bags depends on the time that plants remain in the nursery. If plants retain in the nursery for a long period, a triangular spacing of 75 cm X 75 cm X 75 cm is recommended (figure 9).

❖ Advantages of poly bagged seedlings

1. When seedlings are transplanted after removing the poly bags, field casualties are less and ensure uniform plantation due less root damage at planting.
2. Fast growth, early flowering and bearing.
3. Less field casualties and uniform plantations.
4. Easy to fertilize, control weeds and irrigate.
5. Poly bagged seedlings could be kept for some time until weather conditions are suitable for field planting.



Figure 10: Certified poly bagged seedlings